



## NATURAL LANGUAGE PROCESSING

Project Title

### SEMANTIC SIMILARITY-BASED DUPLICATE QUESTION DETECTION

|                      |                                                                                                                                                            |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programme</b>     | BBA (Artificial Intelligence)                                                                                                                              |
| <b>Academic Year</b> | 2026-27                                                                                                                                                    |
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## 1. Abstract

Question-and-answer platforms accumulate many questions that are worded differently but ask the same thing, which splits answers across several pages. This project addresses duplicate question detection: deciding whether two questions share the same underlying intent. It uses the Quora Question Pairs dataset (404,290 labelled pairs) and compares a TF-IDF with cosine-similarity baseline against Sentence-BERT (all-MiniLM-L6-v2, 384-dimensional embeddings) with cosine similarity. Decision thresholds were selected on validation data (0.17 for TF-IDF and 0.73 for Sentence-BERT). To prevent question-level leakage, unique question IDs were divided into training, validation and test groups and only pairs whose two questions fell in the same group were kept, leaving 198,177, 8,952 and 9,029 pairs respectively. On the test set, Sentence-BERT achieved an accuracy of 0.7649 and an F1-score of 0.7351, against 0.5348 and 0.5940 for TF-IDF. Precision rose from 0.4331 to 0.6184, while recall fell from 0.9449 to 0.9059, so the improvement is not uniform across metrics. Audits by lexical overlap and question length, and an analysis of the 1,817 false positives and 306 false negatives, indicate that Sentence-BERT captures broad semantic similarity but may not preserve fine differences in direction, numbers or dates, and misses some paraphrases that score just below the threshold. Several false positives are near-paraphrases, which points to label ambiguity in the dataset.

## 2. Introduction

### 2.1 Background and context

Online question-and-answer platforms allow users to post questions in free text, and different users often phrase the same need in different ways. When such questions are not linked, answers and readers are divided across several near-identical pages. Identifying duplicate questions is therefore a practical text-matching problem: two questions may share few words yet ask the same thing, or share most of their words yet ask different things.

### 2.2 Motivation

The Quora Question Pairs dataset provides a large set of question pairs with duplicate / non-duplicate labels, which makes it suitable for studying this problem with measurable outcomes. The project was also motivated by a specific question: how far does a pretrained sentence-embedding model improve over a word-overlap baseline, and what kinds of pairs does it still get wrong? For this reason the work does not stop at reporting accuracy; it also audits performance across subgroups and examines individual errors.

### 2.3 Role of NLP

Natural Language Processing supplies the means of representing a question numerically so that two questions can be compared. A lexical representation (TF-IDF) compares the words that two questions share. A semantic representation (Sentence-BERT) maps each question to a dense vector intended to place questions with similar meaning close together, so that similarity can be measured even when the wording differs (Reimers & Gurevych, 2019).

Comparing the two representations on the same data and the same evaluation protocol is the core of this project.

### 3. Problem Statement and Objectives

**Problem statement.** Given a pair of questions, determine whether the two questions have the same underlying meaning or intent (duplicate, label 1) or not (non-duplicate, label 0), using semantic similarity between the questions.

**Objectives.** The project set out to:

1. build a lexical baseline using TF-IDF vectors and cosine similarity;
2. build a semantic model using Sentence-BERT (all-MiniLM-L6-v2) embeddings and cosine similarity;
3. create a leakage-safe training / validation / test split based on question IDs, and verify that no question ID is shared across the splits;
4. select a decision threshold for each method using validation data only, and evaluate both methods once on the held-out test set using accuracy, precision, recall, F1-score and a confusion matrix;
5. audit the Sentence-BERT results across lexical-overlap and question-length subgroups; and
6. analyse representative false positives and false negatives to identify the model's failure patterns.

### 4. Dataset Description

**Dataset.** Quora Question Pairs, obtained from the Kaggle competition page (<https://www.kaggle.com/c/quora-question-pairs/data>). The training file (train.csv) was uploaded to Google Colab and loaded with pandas. The questions inspected in this project are written in English.

**Source:** Quora Question Pairs competition, hosted on Kaggle. The dataset was used for academic project work in accordance with the applicable Kaggle competition terms.

**Size and fields.** The file contains 404,290 question pairs and six columns (Table 1) covering 537,933 unique question IDs. Because the same question can occur in more than one pair, the number of unique question IDs is smaller than the 808,580 question slots implied by 404,290 pairs; this repetition is the source of the leakage risk discussed below.

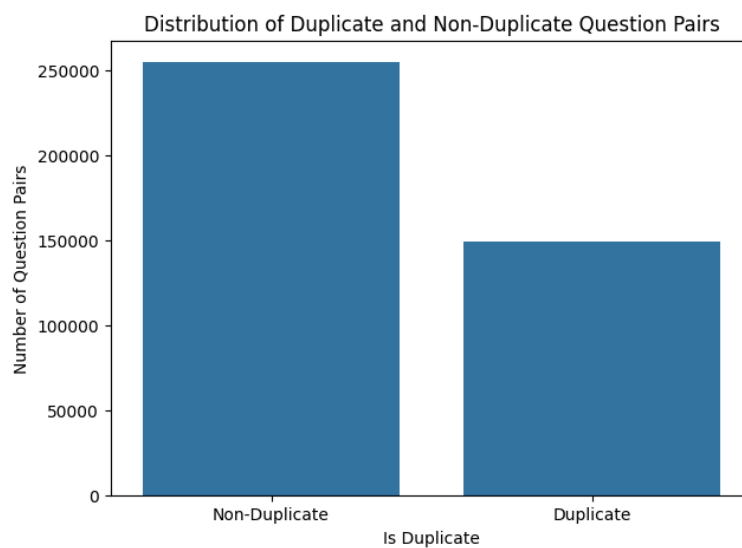
**Table 1:** *Fields of the Quora Question Pairs training file*

| Field | Description                              |
|-------|------------------------------------------|
| id    | Identifier of the question pair          |
| qid1  | Unique identifier of the first question  |
| qid2  | Unique identifier of the second question |

| Field        | Description                                                   |
|--------------|---------------------------------------------------------------|
| question1    | Text of the first question                                    |
| question2    | Text of the second question                                   |
| is_duplicate | Label: 1 = the two questions are duplicates, 0 = they are not |

**Data quality.** One value in question1 and two values in question2 are missing (0.000247% and 0.000495% of rows respectively). No exact duplicate rows were found.

**Class distribution.** The dataset is moderately imbalanced: 255,027 pairs (63.08%) are non-duplicates and 149,263 pairs (36.92%) are duplicates (Figure 1). Accuracy alone is therefore not sufficient to judge a classifier, and precision, recall and F1-score are reported alongside it.



**Figure 1:** Distribution of duplicate and non-duplicate question pairs in the original dataset (404,290 pairs), as produced in the project notebook.

**Other observations.** Duplicate pairs are on average shorter than non-duplicate pairs, and share more words: the mean word-overlap (Jaccard) score was 0.4310 for duplicates and 0.2644 for non-duplicates. Supporting statistics are given in Annexure Table 14.

**Leakage issue and the leakage-safe split.** Splitting the pairs directly at random would allow the same question to appear in the training set in one pair and in the test set in another. To avoid this question-level leakage, the 537,933 unique question IDs (qid1 and qid2 combined) were split at random (random\_state = 42) into 70% training, 15% validation and 15% test groups. A pair was retained only when both of its question IDs belonged to the same group. Pairs whose two questions fell in different groups were excluded.

**Table 2:** Leakage-safe split by question ID and the pairs retained in each split

| Split      | Unique question IDs assigned | Pairs retained | Non-duplicate (%) | Duplicate (%) |
|------------|------------------------------|----------------|-------------------|---------------|
| Training   | 376,553                      | 198,177        | 63.04             | 36.96         |
| Validation | 80,690                       | 8,952          | 63.08             | 36.92         |

| Split        | Unique question IDs assigned | Pairs retained | Non-duplicate (%) | Duplicate (%) |
|--------------|------------------------------|----------------|-------------------|---------------|
| Test         | 80,690                       | 9,029          | 63.99             | 36.01         |
| <b>Total</b> | <b>537,933</b>               | <b>216,158</b> | —                 | —             |

Only 216,158 of the 404,290 original pairs (approximately 53.5%) were retained; the other 188,132 pairs were excluded because their two questions belonged to different groups. All experiments therefore use this retained subset and not the full dataset. The question-ID sets of the three splits were checked and found to be disjoint:  $\text{Train} \cap \text{Validation} = 0$ ,  $\text{Train} \cap \text{Test} = 0$  and  $\text{Validation} \cap \text{Test} = 0$ .

## 5. Data Preprocessing

Preprocessing was deliberately minimal. Only the steps below were carried out in the notebook.

- **Handling missing text.** Missing values in question1 and question2 (one and two rows in the original file) were replaced with empty strings before vectorisation and word counting, so that every pair could be processed.
- **Lowercasing, stop-word removal and n-grams (TF-IDF branch only).** These were performed inside the TF-IDF vectoriser: text was lowercased, English stop words were removed, and unigrams and bigrams were extracted using scikit-learn’s default tokenisation. Terms occurring in fewer than two training questions were ignored and the vocabulary was capped at 100,000 terms.
- **Sentence-BERT branch.** The question text was passed to the model without additional cleaning; tokenisation was handled by the model’s own tokenizer. The resulting embeddings were L2-normalised.
- **Audit features.** For the lexical-overlap audit, each question was lowercased and split on whitespace, and overlap was measured as the Jaccard score (shared words divided by total distinct words). Question length was measured as the number of whitespace-separated words.

No stemming, lemmatisation, spelling correction, punctuation stripping, or special handling of URLs, hashtags, mentions or emojis was applied, and no multilingual or code-mixed processing was needed or performed. Rows with missing text were not removed. Test-set errors in Section 11 should be read with this in mind: differences in punctuation, contractions and capitalisation were left in the text that the models received (although the TF-IDF vectoriser lowercases its input).

## 6. Feature Representation / Embeddings

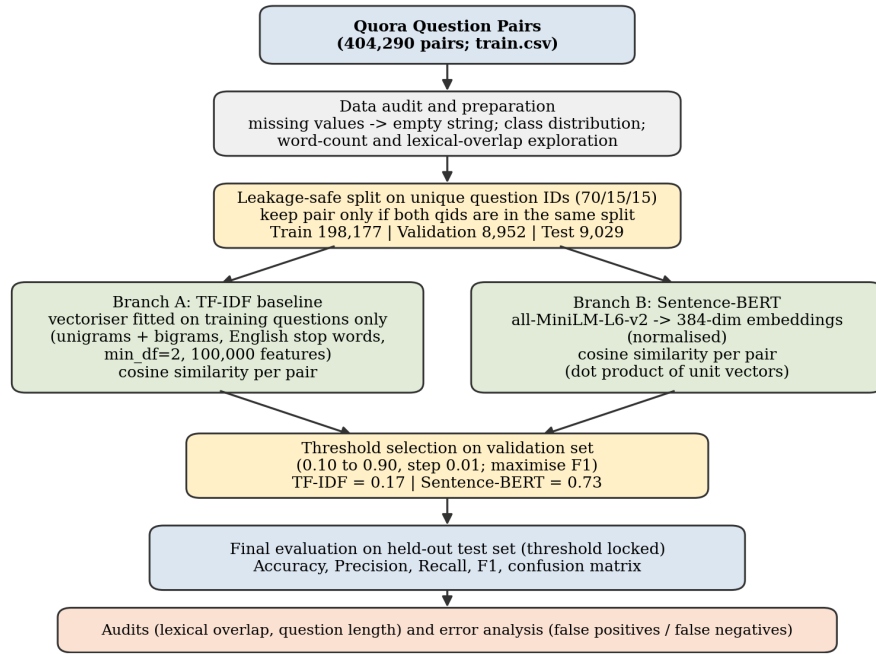
Two representations were used so that a lexical view and a semantic view of the same questions could be compared.

**TF-IDF.** Term Frequency–Inverse Document Frequency represents each question as a sparse vector in which each term is weighted by how often it occurs in the question and down-weighted if it is common across the training questions. Two questions are then compared by the cosine of the angle between their vectors, which is high when they share distinctive terms. TF-IDF was chosen as the baseline because it is simple, interpretable and depends only on word overlap. Its limitation is that it cannot recognise a paraphrase that uses different words. The vectoriser was fitted on the training questions only (question1 and question2 pooled), with unigrams and bigrams, English stop-word removal, a minimum document frequency of 2 and at most 100,000 features. The fitted vocabulary reached the 100,000-term limit.

**Sentence-BERT.** Sentence-BERT (Reimers & Gurevych, 2019) adapts BERT-style networks to produce a fixed-size embedding for a whole sentence, so that sentences can be compared directly with cosine similarity. This project used the pretrained model all-MiniLM-L6-v2 from the Sentence Transformers library, which outputs 384-dimensional embeddings. It was chosen because it is designed for sentence-level comparison, is small enough to run in Google Colab, and can represent questions with similar meaning but different wording. Embeddings were normalised to unit length, so the cosine similarity of two questions equals the dot product of their embeddings. The model was used as provided and was not fine-tuned on this dataset.

## 7. Proposed Methodology / System Architecture

The end-to-end workflow is shown in Figure 2. Both methods follow the same protocol: the question pair is converted to two vectors, a cosine similarity score is computed, and the pair is classified as a duplicate if the score is at or above a threshold chosen on the validation set. The audits and error analysis were carried out on the Sentence-BERT test predictions.



**Figure 2:** System workflow of the duplicate question detection project, from the input dataset to evaluation, audits and error analysis.

The workflow consists of the following stages:

1. **Input.** Question pairs with duplicate labels are read from train.csv.
2. **Preparation.** Missing text is replaced with empty strings; the class distribution, question length and lexical overlap are examined.
3. **Leakage-safe split.** Unique question IDs are divided 70/15/15 and pairs are kept only when both questions lie in the same split.
4. **Representation and similarity.** TF-IDF vectors (fitted on training questions) and Sentence-BERT embeddings are produced for the validation and test questions, and a cosine similarity is computed for each pair.
5. **Thresholding.** For each method, the threshold between 0.10 and 0.90 (step 0.01) that gives the highest validation F1-score is selected and then fixed.
6. **Output and evaluation.** Each test pair is labelled duplicate or non-duplicate and evaluated with accuracy, precision, recall, F1-score and a confusion matrix, followed by subgroup audits and error analysis.

## 8. NLP Model / Technique Used

Both approaches are similarity-based classifiers: neither learns a classification layer from the labels. The only quantities fitted from the data are the TF-IDF vocabulary and weights (training questions) and the decision threshold (validation pairs). Table 3 summarises the settings.

**Table 3:** Configuration of the two methods compared

| Setting         | TF-IDF baseline                                                                   | Sentence-BERT                                                        |
|-----------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Representation  | Sparse TF-IDF vectors                                                             | 384-dimensional dense embeddings                                     |
| Model / library | scikit-learn TfidfVectorizer                                                      | Sentence Transformers, all-MiniLM-L6-v2 (pretrained, not fine-tuned) |
| Key settings    | Lowercase; English stop words; n-grams (1, 2); min_df = 2; max_features = 100,000 | Normalised embeddings; batch size 64 for encoding                    |
| Fitted on       | Training questions only                                                           | No fitting (pretrained)                                              |
| Similarity      | Cosine similarity                                                                 | Cosine similarity (dot product of unit vectors)                      |
| Decision rule   | Duplicate if similarity $\geq 0.17$                                               | Duplicate if similarity $\geq 0.73$                                  |

**Threshold selection.** Thresholds from 0.10 to 0.90 in steps of 0.01 were evaluated on the validation pairs, and the one giving the highest F1-score was chosen: 0.17 for TF-IDF and 0.73 for Sentence-BERT (Table 4). The thresholds were then fixed and applied unchanged to the test set. Because F1 was the selection criterion, the thresholds were not tuned for accuracy; at both selected thresholds recall is higher than precision.

**Table 4:** Validation-set performance at the selected thresholds (8,952 pairs)

| Method        | Threshold | Val. accuracy | Val. precision / recall | Val. F1 |
|---------------|-----------|---------------|-------------------------|---------|
| TF-IDF        | 0.17      | 0.5412        | 0.4428 / 0.9398         | 0.6020  |
| Sentence-BERT | 0.73      | 0.7530        | 0.6129 / 0.8986         | 0.7287  |

**Evaluation approach.** The duplicate class (label 1) is treated as the positive class. Accuracy is the share of all pairs classified correctly; precision is the share of predicted duplicates that are true duplicates; recall is the share of true duplicates that are found; and F1 is the harmonic mean of precision and recall. The confusion matrix reports true negatives (TN), false positives (FP), false negatives (FN) and true positives (TP). Test data were used only for the final evaluation, the subgroup audits and the error analysis.

## 9. Implementation

The project was implemented in Python in a Google Colab notebook (Semantic\_Similarity\_Duplicate\_Question\_Detection\_CA3.ipynb). The main libraries were

pandas and NumPy (data handling), matplotlib and seaborn (visualisation), scikit-learn (TfidfVectorizer, cosine similarity, data splitting and evaluation metrics) and Sentence Transformers (version 5.7.0 in the final run) for the embedding model. The notebook is organised into the modules listed in Table 5; the complete code is kept in the notebook and repository rather than reproduced here.

**Table 5:** *Major modules of the notebook implementation*

| Module                      | What it does                                                                                                                                               |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data loading and inspection | Loads train.csv; checks size, data types, missing values, exact duplicates and class distribution                                                          |
| Exploratory analysis        | Computes word and character counts and Jaccard lexical overlap; plots class distribution and overlap by class                                              |
| Leakage-safe splitting      | Splits unique question IDs 70/15/15, builds the three pair sets, and checks that question-ID sets do not overlap                                           |
| TF-IDF baseline             | Fits the vectoriser on training questions, computes pair similarity, selects the threshold on validation data and evaluates on test data                   |
| Sentence-BERT model         | Encodes the unique validation and test questions (15,543 and 15,964 questions), computes pair similarity, selects the threshold and evaluates on test data |
| Audits                      | Evaluates Sentence-BERT on low/high lexical-overlap and short/long question subgroups using median cut-offs                                                |
| Error analysis              | Labels each test prediction as TP, TN, FP or FN and lists the highest-scoring false positives and false negatives                                          |
| Visualisation               | Produces the model-comparison chart and the Sentence-BERT confusion matrix                                                                                 |

No separate application or user interface was developed; the notebook is the working system, and its outputs are presented as figures and tables in this report. Because the Sentence-BERT embeddings were computed once per unique question and then looked up for each pair, repeated questions were not encoded more than once.

## 10. Results and Evaluation

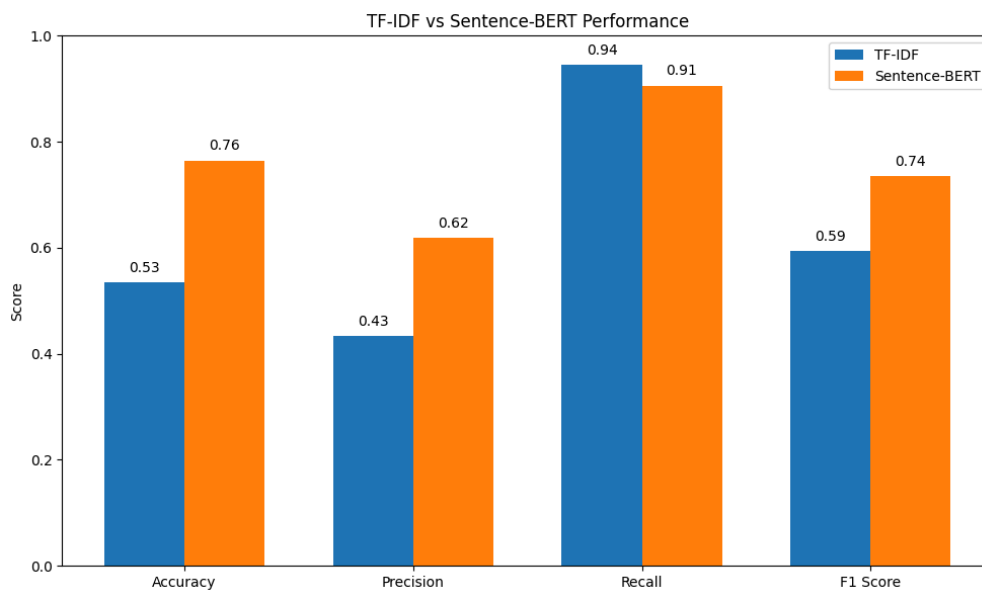
### 10.1 Final test results

Table 6 and Figure 3 give the final test results on the 9,029 held-out pairs, evaluated at the validation-selected thresholds.

**Table 6:** *Final test-set results: TF-IDF baseline versus Sentence-BERT*

| Model                             | Accuracy      | Precision     | Recall       | F1-score      |
|-----------------------------------|---------------|---------------|--------------|---------------|
| TF-IDF baseline                   | 0.5348        | 0.4331        | 0.9449       | 0.5940        |
| Sentence-BERT                     | 0.7649        | 0.6184        | 0.9059       | 0.7351        |
| <b>Change (percentage points)</b> | <b>+23.00</b> | <b>+18.53</b> | <b>−3.91</b> | <b>+14.11</b> |

*Note: the change row is computed from unrounded scores, so it can differ by 0.01 from the difference of the rounded values shown above.*



**Figure 3:** Comparison of TF-IDF and Sentence-BERT on the test set across accuracy, precision, recall and F1-score, as produced in the project notebook.

Sentence-BERT substantially improved overall classification quality: accuracy rose from 53.48% to 76.49% and F1-score from 59.40% to 73.51%, while duplicate-detection recall remained high at 90.59%. The improvement is not uniform. Precision increased by 18.53 percentage points, but recall decreased by 3.91 percentage points, so Sentence-BERT did not improve every metric. In practical terms, Sentence-BERT found slightly fewer of the true duplicates than TF-IDF, but was far more reliable when it predicted a duplicate.

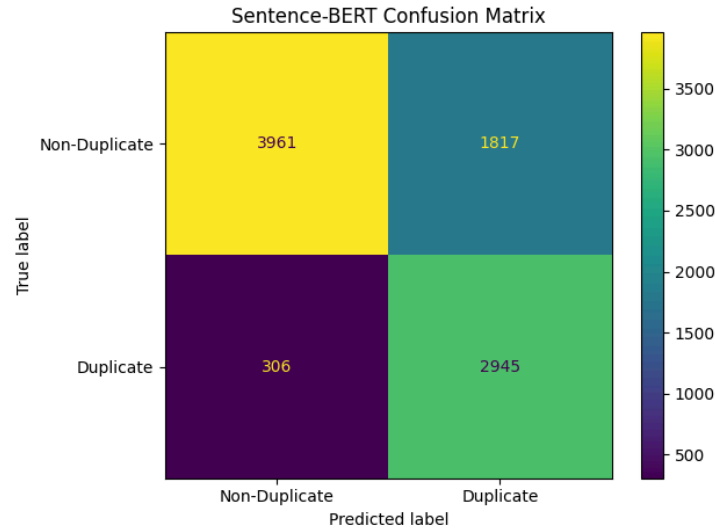
For context, 63.99% of the test pairs are non-duplicates (5,778 of 9,029, from the confusion matrix). A classifier that labelled every pair a non-duplicate would therefore reach 63.99% accuracy, which is higher than the TF-IDF accuracy of 53.48% and lower than the Sentence-BERT accuracy of 76.49%. This reference figure is derived from the class counts and was not a model run in the notebook. It shows why the TF-IDF baseline at its F1-selected threshold is weak: it predicts most pairs as duplicates (7,093 of 9,029), which gives high recall but low precision.

## 10.2 Confusion matrices

Table 7 lists the confusion-matrix counts of both methods, and Figure 4 shows the Sentence-BERT confusion matrix.

**Table 7:** Test-set confusion-matrix counts (duplicate = positive class)

| Model           | TN    | FP    | FN  | TP    |
|-----------------|-------|-------|-----|-------|
| TF-IDF baseline | 1,757 | 4,021 | 179 | 3,072 |
| Sentence-BERT   | 3,961 | 1,817 | 306 | 2,945 |



**Figure 4:** Confusion matrix of the Sentence-BERT model on the test set (threshold 0.73).

Of the 5,778 true non-duplicates, Sentence-BERT correctly rejected 3,961 and wrongly accepted 1,817 as duplicates; of the 3,251 true duplicates, it found 2,945 and missed 306. Compared with TF-IDF, Sentence-BERT produced 2,204 fewer false positives (1,817 against 4,021) at the cost of 127 more false negatives (306 against 179). False positives remain the larger error type for Sentence-BERT.

### 10.3 Audit 1: lexical overlap

To test whether performance depends on how many words the two questions share, the test pairs were split at the median lexical-overlap (Jaccard) score of 0.2727 into a low-overlap group (below the median) and a high-overlap group (at or above it). The audit was applied to the Sentence-BERT predictions (Table 8).

**Table 8:** Sentence-BERT test performance by lexical-overlap and question-length subgroups

| Subgroup             | Pairs | Accuracy | Precision | Recall | F1-score |
|----------------------|-------|----------|-----------|--------|----------|
| Low lexical overlap  | 4,446 | 0.8561   | 0.5647    | 0.7928 | 0.6596   |
| High lexical overlap | 4,583 | 0.6764   | 0.6346    | 0.9417 | 0.7582   |
| Short question pairs | 4,345 | 0.7381   | 0.6378    | 0.9416 | 0.7605   |
| Long question pairs  | 4,684 | 0.7897   | 0.5899    | 0.8544 | 0.6979   |

High-overlap pairs had substantially higher recall (0.9417 against 0.7928) but lower accuracy (0.6764 against 0.8561). This is consistent with the model being more likely to classify highly overlapping pairs as duplicates, which would also increase false positives. Subgroup-level confusion matrices were not produced, so this is an interpretation rather than a measured count, and the audit does not establish that overlap causes the difference.

### 10.4 Audit 2: question length

Pair length was defined as the average word count of the two questions. The median was 10.0 words; pairs below the median were treated as short and pairs at or above it as long (Table 8,

lower rows). Short pairs achieved higher recall (0.9416 against 0.8544) and F1-score (0.7605 against 0.6979), whereas long pairs achieved higher accuracy (0.7897 against 0.7381) but lower recall and F1-score. Precision was higher for short pairs (0.6378 against 0.5899). The two length groups therefore trade off differently across metrics, and neither is uniformly better.

## 11. Error Analysis / NLP Failure Analysis

Of the 9,029 test pairs, Sentence-BERT made 1,817 false-positive and 306 false-negative errors. The notebook sorted each error type by similarity score and printed the ten highest-scoring examples; five of each are shown below (Tables 9 and 10) and the full lists of ten are in Annexure Tables 12 and 13. Because they are the extreme cases of each error type, they illustrate failure patterns but do not measure how common each pattern is across all errors.

### 11.1 False positives

False positives are non-duplicate pairs (label 0) with a similarity score at or above 0.73. The ten highest-scoring ones had scores between 0.9904 and 0.9991.

**Table 9:** Selected false positives (actual label = 0) with Sentence-BERT similarity; Examples 1, 5, 7, 8 and 10 of the ten highest-scoring cases

| No. | Question 1                                                                                                        | Question 2                                                                                                           | Similarity |
|-----|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------|
| 1   | How do I convince youth to celebrate Indian festivals in the traditional way?                                     | How do I convince youth to celebrate Indian festivals in traditional way?                                            | 0.9991     |
| 5   | How can I switch my phone from Verizon to Straight Talk?                                                          | How can I switch my phone from straight talk to Verizon?                                                             | 0.9950     |
| 7   | How can I solve this question of Aldehydes and Ketones?                                                           | How can you solve this question of aldehydes and ketones?                                                            | 0.9918     |
| 8   | What business can one run from a laptop and make around £1 million a year with less than £2,000 starting capital? | What business can one run from a laptop and make more than £1 million a year with less than £5,000 starting capital? | 0.9908     |
| 10  | When would be the next batch of Infosys after November 21st?                                                      | When would be the next batch of Infosys after November 7th?                                                          | 0.9904     |

Two kinds of case appear among these ten examples. In the first, the pairs differ in details that change the meaning of the question. Example 5 reverses the direction of the switch (Verizon to Straight Talk against Straight Talk to Verizon), Example 8 changes the numerical constraints (around against more than £1 million; £2,000 against £5,000 starting capital) and Example 10 changes the date (November 21st against November 7th). Sentence-BERT gave these pairs scores above 0.99, so its embedding-based similarity did not preserve these distinctions. One possible explanation is that the sentence embedding captures the overall semantic context strongly enough that changes in a swapped entity, numerical value or date may have limited effect on the resulting similarity score. This explanation was not directly tested in the present study.

In the second kind of case, the two questions differ only in an article, a contraction, punctuation or a minor word (for example, “in the traditional way” against “in traditional way”, or “How do I” against “How can I” in Example 9 of Annexure Table 12). Seven of the ten highest-scoring false positives are of this kind. Many of these read as paraphrases even though the dataset labels them non-duplicate. These errors therefore should not all be treated as genuine model failures; some reflect label ambiguity, i.e. pairs that are hard to distinguish semantically and where the label itself is debatable. Whether a given label is correct was not verified in this project.

## 11.2 False negatives

False negatives are duplicate pairs (label 1) with a similarity score below 0.73. The ten highest-scoring ones all had scores between 0.7281 and 0.7299.

**Table 10:** Selected false negatives (actual label = 1) with Sentence-BERT similarity; Examples 1, 4, 5, 8 and 10 of the ten highest-scoring cases

| No. | Question 1                                                                                    | Question 2                                                                             | Similarity |
|-----|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------|
| 1   | How do I get rid of scalp acne?                                                               | How do I to get rid of acne scars?                                                     | 0.7299     |
| 4   | How do I get rid of lethargy?                                                                 | What are some good ways to get rid of weekend lethargy?                                | 0.7296     |
| 5   | How do you get rid of a addiction?                                                            | What is the best way to get rid of addictions like Facebook and WhatsApp?              | 0.7295     |
| 8   | What are the risks of smoking weed (ganja)?<br>Does it cause cancer?                          | Does weed give you lung cancer?                                                        | 0.7289     |
| 10  | What are some new security features introduced in the new ₹2000 and ₹500 bills issued by RBI? | What are some interesting features of the new 500 and 2000 rupees notes issued by RBI? | 0.7281     |

These ten pairs sit within 0.002 of the selected threshold of 0.73, which shows how sensitive the classification is to the exact threshold value. Because the list is sorted by score, these are by construction the false negatives closest to the boundary; the notebook did not examine the score distribution of all 306 false negatives, and the effect of lowering the threshold on false positives was not evaluated. Most of these pairs are plausible duplicates that differ in specificity or wording: in Example 4 one question asks about lethargy in general and the other about weekend lethargy, and in Example 5 one asks about addiction in general and the other about addictions such as Facebook and WhatsApp. Some labels are also debatable; the first pair in Table 10 (scalp acne against acne scars) asks about arguably different conditions yet is labelled a duplicate. These cases again suggest that part of the error comes from ambiguity in what counts as a duplicate.

## 11.3 Summary of failure patterns

Sentence-BERT captures broad semantic similarity effectively, but its embedding-based similarity score may not sufficiently preserve fine-grained distinctions involving direction, numerical values, dates, entities or contextual constraints. Conversely, some duplicate paraphrases receive scores just below the selected threshold and are therefore classified as non-duplicates. In addition, the near-paraphrase false positives and the debatable false

negatives show that some of the disagreement between model and label may reflect ambiguity in the dataset labels rather than a clear model error.

## 12. Ethical Considerations and Limitations

### 12.1 Ethical considerations

- **Consequences of wrong merging.** In a deployed setting, a false positive could cause two different questions to be merged or hidden, and a false negative would leave duplicates unlinked. With 1,817 false positives on 9,029 test pairs, automatic merging without human review would not be appropriate on the basis of this project's results.
- **Content of the data.** The dataset consists of user-written questions on many subjects, some of which concern sensitive topics (for example, the test errors include a question about being gay in Hong Kong). The data were used as released on Kaggle for academic analysis; no attempt was made to identify or profile the question authors.
- **Fairness.** Performance was not analysed across topics, communities or demographic groups, so possible uneven performance on particular subject areas has not been assessed.

### 12.2 Dataset limitations

- Only 216,158 of 404,290 pairs (about 53.5%) were retained by the leakage-safe split, so the results describe this subset. The test set's duplicate share (36.01%) differs slightly from the original dataset (36.92%).
- Some labels appear ambiguous, as seen in near-paraphrase false positives and in debatable duplicate labels (Section 11). This puts a ceiling on how much of the measured error can be attributed to the model.
- The questions inspected were in English, so the findings should not be assumed to hold for other languages.

### 12.3 Evaluation and model limitations

- Results come from one random split (`random_state = 42`) and one run; no repeated splits, cross-validation or confidence intervals were computed, so small differences should be interpreted cautiously.
- Thresholds were selected on a validation set of 8,952 pairs by maximising F1, and the resulting operating point favours recall over precision for both methods.
- Only one pretrained Sentence-BERT model was tested, without fine-tuning, and no stronger or alternative models were compared. The TF-IDF baseline is a simple one and its performance at its selected threshold is close to what would be obtained by labelling most pairs as duplicates.
- The audits used median cut-offs on two features (lexical overlap and length) and did not include subgroup confusion matrices. The error analysis is based on the ten highest-scoring examples of each error type rather than a random sample.

- The model’s similarity score may not preserve fine-grained differences in direction, numbers, dates, entities or constraints, and can place some duplicate paraphrases just below the threshold (Section 11).

## 13. Conclusion

This project built and compared two duplicate-question detectors on the Quora Question Pairs data: a TF-IDF baseline and a pretrained Sentence-BERT model, both using cosine similarity with a validation-selected threshold. A question-ID-based split was used to prevent leakage, and the question-ID sets of the training, validation and test groups were verified to be disjoint. The cost of this safeguard is that only 216,158 of the 404,290 original pairs were retained.

On the held-out test set, Sentence-BERT raised accuracy from 53.48% to 76.49% and F1-score from 59.40% to 73.51%, and precision from 43.31% to 61.84%, while recall decreased from 94.49% to 90.59%. Sentence-BERT therefore improved overall classification quality while maintaining high duplicate-detection recall, but it did not improve every metric. The audits showed that performance varies with lexical overlap and question length, and the error analysis showed two failure patterns: high similarity scores for pairs that differ in fine-grained details, and duplicate paraphrases scoring just below the threshold. Some errors also reflect label ambiguity.

All six objectives in Section 3 were achieved: both models were built and evaluated on a leakage-safe split with validation-selected thresholds, the Sentence-BERT results were audited, and representative errors were analysed. The main learning from the project is that a single accuracy figure hides important trade-offs, and that subgroup audits and example-level error analysis are needed to understand what a model gets right and wrong.

## 14. Future Scope

The following extensions are realistic next steps; none of them was carried out in this project.

- Fine-tune a sentence-embedding model on the training pairs, and compare it against the pretrained model and against other embedding models.
- Add a pair-level classifier or cross-encoder stage that sees both questions together, which may help with differences in direction, numbers, dates and entities.
- Add explicit checks or features for numbers, dates and named entities, and examine whether they reduce false positives such as those in Section 11.
- Study threshold sensitivity more fully, for example by plotting precision and recall against the threshold and by testing thresholds chosen for different goals (e.g. higher precision before automatic merging).
- Repeat the experiments over several random splits and report confidence intervals; also evaluate on the pairs excluded by the leakage-safe split in a separate, clearly labelled analysis.

- Review a random sample of false positives and false negatives manually to estimate how many reflect label ambiguity, and analyse performance by topic.

## 15. References

References follow the APA author–date style. Publication years are given only where they are known.

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## 16. GitHub Repository

GitHub Repository Link:

[\[https://github.com/priyanilekhi/semantic-similarity-duplicate-question-detection\]](https://github.com/priyanilekhi/semantic-similarity-duplicate-question-detection)

The repository contains the project notebook  
(Semantic\_Similarity\_Duplicate\_Question\_Detection\_CA3.ipynb) and the files required to understand the project.

## 17. Team Contribution

**Table 11:** *Contribution of each team member*

| Team Member     | PRN         | Specific Contribution                                                                                                                                                           |
|-----------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priyani Lekhi   | 25030422093 | Overall Project Head - handled the project from problem statement and conceptualisation through implementation coordination, analysis, audit, report direction and final pitch. |
| Kudrat Thakkar  | 25030422062 | Code implementation and report development.                                                                                                                                     |
| Bhavay Madan    | 25030422032 | PPT development and pitch presentation.                                                                                                                                         |
| Dhruvi Aggarwal | 25030422039 | Dataset work and pitch presentation.                                                                                                                                            |

## 18. Annexure

The annexure contains supporting material that is useful but not essential to the main report.

### A.1 Full list of the ten highest-scoring false positives

**Table 12:** Ten highest-scoring Sentence-BERT false positives on the test set (actual label = 0)

| No. | Question 1                                                                                                        | Question 2                                                                                                           | Similarity |
|-----|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------|
| 1   | How do I convince youth to celebrate Indian festivals in the traditional way?                                     | How do I convince youth to celebrate Indian festivals in traditional way?                                            | 0.9991     |
| 2   | Is human interaction possible in an absolute silence?                                                             | Is human interaction possible in absolute silence?                                                                   | 0.9960     |
| 3   | What is it like to be gay in Hong Kong?                                                                           | What's it like to be gay in Hong Kong?                                                                               | 0.9959     |
| 4   | What is the single most effective piece of financial advice you've ever received?                                 | What is the single, most effective piece of financial advice you've ever received?                                   | 0.9950     |
| 5   | How can I switch my phone from Verizon to Straight Talk?                                                          | How can I switch my phone from straight talk to Verizon?                                                             | 0.9950     |
| 6   | What is the english translation for the word in the image?                                                        | What is the english translation of the word in the image?                                                            | 0.9939     |
| 7   | How can I solve this question of Aldehydes and Ketones?                                                           | How can you solve this question of aldehydes and ketones?                                                            | 0.9918     |
| 8   | What business can one run from a laptop and make around £1 million a year with less than £2,000 starting capital? | What business can one run from a laptop and make more than £1 million a year with less than £5,000 starting capital? | 0.9908     |
| 9   | How do I get muscle mass?                                                                                         | How can I get muscle mass?                                                                                           | 0.9908     |
| 10  | When would be the next batch of Infosys after November 21st?                                                      | When would be the next batch of Infosys after November 7th?                                                          | 0.9904     |

### A.2 Full list of the ten highest-scoring false negatives

**Table 13:** Ten highest-scoring Sentence-BERT false negatives on the test set (actual label = 1)

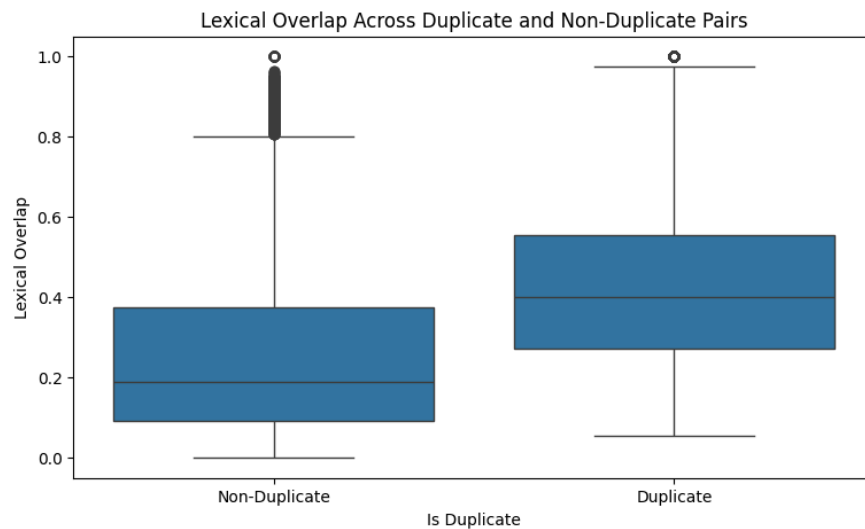
| No. | Question 1                                                                 | Question 2                                                                                                                     | Similarity |
|-----|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------|
| 1   | How do I get rid of scalp acne?                                            | How do I to get rid of acne scars?                                                                                             | 0.7299     |
| 2   | What are the high level procedural programming languages?                  | What is the difference between a procedural language and a programming language?                                               | 0.7299     |
| 3   | Career Advice: What are the success tricks for preparing Gate in 3 months? | What is the best way to prepare for gate 2016 (EE) in 3 months?                                                                | 0.7297     |
| 4   | How do I get rid of lethargy?                                              | What are some good ways to get rid of weekend lethargy?                                                                        | 0.7296     |
| 5   | How do you get rid of a addiction?                                         | What is the best way to get rid of addictions like Facebook and WhatsApp?                                                      | 0.7295     |
| 6   | What problems did you face after the ban on 500 and 1000 rupee notes?      | What are all the possible problems that a common Indian citizen may face after scraping the 500 and 1000 rupee note overnight? | 0.7293     |
| 7   | How do I start conversation with a girl at my workspace?                   | What are some of the best ways to start a conversation with a girl?                                                            | 0.7292     |
| 8   | What are the risks of smoking weed (ganja)? Does it cause cancer?          | Does weed give you lung cancer?                                                                                                | 0.7289     |

| No. | Question 1                                                                                    | Question 2                                                                             | Similarity |
|-----|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------|
| 9   | How do I get my dachshund to stop chewing furniture?                                          | How do I stop my puppy from chewing my shoes?                                          | 0.7289     |
| 10  | What are some new security features introduced in the new ₹2000 and ₹500 bills issued by RBI? | What are some interesting features of the new 500 and 2000 rupees notes issued by RBI? | 0.7281     |

### A.3 Supporting dataset statistics

**Table 14:** Word-count and lexical-overlap statistics of the original 404,290 pairs

| Statistic (original dataset)             | All pairs     | Non-duplicate | Duplicate |
|------------------------------------------|---------------|---------------|-----------|
| Mean words in question1                  | 10.94         | 11.58         | 9.85      |
| Mean words in question2                  | 11.18         | 11.96         | 9.86      |
| Mean characters in question1 / question2 | 59.54 / 60.11 | —             | —         |
| Mean lexical overlap (Jaccard)           | —             | 0.2644        | 0.4310    |



**Figure 5:** Box plot of lexical overlap for non-duplicate and duplicate pairs in the original dataset, as produced in the project notebook.

The box plot (Figure 5) is consistent with the mean values in Table 14: duplicate pairs tend to share more words than non-duplicate pairs, although the two classes overlap, which is why lexical overlap alone is not a sufficient signal.